EgoLife: Towards Egocentric Life Assistant

Anonymous Authors
Anonymous Institution
anonymous@email.com

Abstract

We present EgoLife, the first week-long, multimodal egocentric dataset featuring 266 hours of continuous recordings from 6 participants over 7 days in a shared living environment. Unlike existing egocentric datasets that capture short, isolated activities, EgoLife provides ultra-long-duration recordings essential for developing practical life assistants. We introduce EgoLifeQA, a comprehensive benchmark with 3,000 multiple-choice questions spanning five key categories: EntityLog, EventRecall, HabitInsight, RelationMap, and TaskMaster. Our analysis reveals that 67% of questions require more than 2 hours of context, highlighting the critical need for long-horizon reasoning. To address this challenge, we propose EgoButler, a novel framework combining EgoGPT for omni-modal understanding with EgoRAG for hierarchical memory retrieval. EgoGPT, trained on our curated EgoIT-99K instruction dataset, achieves state-of-the-art performance on existing egocentric benchmarks (75.4 on EgoSchema, 61.4 on EgoThink, 33.4 on EgoPlan). EgoRAG significantly improves long-context reasoning, boosting performance on questions requiring ¿24h context from 25.0 to 35.4. Our work establishes a foundation for developing practical egocentric life assistants capable of ultra-long-horizon reasoning.

1. Introduction

Imagine a personal AI assistant that remembers your daily routines, tracks your habits, and provides contextualized help based on weeks of observations. While this vision of an egocentric life assistant has captured both research and public imagination, existing datasets and methods fall short of the requirements for such systems. Current egocentric datasets like Ego4D [1] and EgoExo4D [2] focus on short, task-specific activities ranging from minutes to hours, missing the long-term behavioral patterns and social dynamics essential for life assistance scenarios.

The core challenge lies in the temporal scope: real-life assistance requires understanding patterns that emerge over days or weeks, not just individual actions captured in short clips. Consider questions like "What breakfast did I have three days ago?" or "How has my sleep schedule changed this week?" These queries demand access to ultra-long-context information spanning multiple days, far beyond the capabilities of current egocentric benchmarks and models.

In this work, we introduce **EgoLife**, the first week-long, multimodal egocentric dataset specifically designed for life assistant research. Our dataset captures 266 hours of continuous recordings from 6 participants over 7 consecutive days in a shared living environment, instrumented with Meta Aria glasses, exocentric cameras, and mmWave sensors. This rich multimodal setup enables comprehensive understanding of both individual behaviors and interpersonal dynamics.

Building on EgoLife, we present **EgoLifeQA**, a carefully designed benchmark featuring 3,000 multiple-choice questions across five categories that mirror real-world assistance scenarios: EntityLog (tracking objects and locations), EventRecall (remembering specific events), HabitInsight (understanding behavioral patterns), RelationMap (modeling social interactions), and TaskMaster (activity planning and management). Crucially, our analysis reveals that 2,003 out of 3,000 questions (67%) require more than 2 hours of context, with many needing access to information spanning multiple days.

To address the unique challenges of ultra-long-context egocentric reasoning, we propose **EgoButler**, a two-stage framework combining clip-level understanding with hierarchical memory retrieval. The first component, EgoGPT, is an omni-modal model extending LLaVA-OneVision [3] with audio capabilities via Whisper v3, trained on our curated EgoIT-99K instruction dataset spanning 9 classic egocentric datasets. The second component, EgoRAG, implements hierarchical retrieval over multi-level memory banks (clip/hour/-day summaries) to enable long-context question answering.

Our contributions are threefold: (1) We present EgoLife, the first week-long multimodal egocentric dataset with dense annotations enabling life assistant research; (2) We introduce EgoLifeQA, a comprehensive benchmark highlighting the importance of ultra-long context for egocentric reasoning; (3) We propose EgoButler, demonstrating significant improvements on long-context questions through omni-modal under-

standing and hierarchical retrieval. EgoGPT achieves state-of-the-art results on existing benchmarks, while EgoRAG provides substantial gains on questions requiring extended temporal context, improving performance on ¿24h queries from 25.0 to 35.4.

Our work opens new research directions toward practical egocentric life assistants, bridging the gap between short-term action recognition and long-term behavioral understanding.

2. Related Work

Egocentric Video Datasets. The field of egocentric video understanding has been significantly advanced by large-scale datasets. EPIC-KITCHENS [4] pioneered kitchen activity recognition with 100 hours of unscripted cooking videos. Ego4D [1] dramatically expanded the scope with 3,670 hours of daily life activities across diverse scenarios, establishing numerous benchmark tasks for first-person perception. EgoExo4D [2] introduced simultaneous egocentric and exocentric capture for skilled activities, enabling new research directions in multi-view understanding.

However, as shown in Table 1, existing datasets focus on relatively short video segments (8.5–22.8 minutes per clip) and lack the ultra-long temporal continuity needed for life assistant applications.

While Ego4D contains extensive footage, its segmented nature misses the longitudinal behavioral patterns essential for understanding personal habits and routines. Moreover, most datasets lack interpersonal dynamics, focusing primarily on individual activities.

Egocentric Video Understanding Models. Recent advances in multimodal large language models have shown promising results on egocentric benchmarks. Commercial models like GPT-4V [7] and Gemini-1.5-Pro [8] achieve strong performance but lack audio processing capabilities crucial for egocentric understanding. Open-source models including LLaVA-OneVision [3], Qwen2-VL [9], and video-specific variants like LLaVA-Next-Video [10] have made significant progress on vision-language tasks.

However, existing models typically process short video clips (30–60 seconds) and struggle with ultra-long context reasoning required for life assistance. While some recent work explores longer video understanding [11], none address the week-long temporal scales necessary for practical life assistants.

Long-Context Video Understanding. The challenge of long-context video understanding has received increasing attention. Approaches include memory-based architectures, hierarchical summarization, and retrieval-augmented generation. However, most work focuses on extending context from minutes to hours, not the days-to-weeks scale required for life assistance scenarios. Our EgoRAG component specifically addresses this gap through hierarchical memory organization and efficient retrieval mechanisms.

Personal AI Assistants. While commercial personal assistants like Siri and Alexa provide basic functionality, they lack the visual and contextual understanding needed for comprehensive life assistance. Recent research in embodied AI and personal robotics has explored visual memory systems, but these typically operate on short temporal scales. Our work represents a significant step toward assistants capable of understanding long-term behavioral patterns and providing contextualized support based on extended observation.

3. The EgoLife Dataset & Benchmark

3.1. EgoLife Dataset

Data Collection Setup. We collected the EgoLife dataset in a carefully instrumented shared living environment we call "EgoHouse". Our multimodal sensing setup includes: (1) Meta Aria glasses worn by participants, capturing egocentric video (1408×1408 at 30fps), audio, IMU, and eye-gaze data; (2) 15 statically mounted GoPro cameras providing exocentric views of common areas; and (3) 2 mmWave radar sensors on the second floor for motion tracking. Figure 2 shows the 3D reconstruction of the house with sensor placements.

Data Collection Protocol. Six participants (3 males, 3 females) cohabited in the EgoHouse for 7 consecutive days, following their normal daily routines including work, meals, leisure activities, and social interactions. Participants wore Aria glasses for approximately 8 hours per day, resulting in 266 total hours of multimodal recordings. Data was synchronized using our EgoSync protocol, ensuring temporal alignment across all sensor modalities.

Privacy and Ethics. Following rigorous privacy protection protocols, we applied EgoBlur to automatically mask faces and personally identifiable information. All participants provided informed consent, and the study was approved by our institutional review board. Recordings were conducted primarily in Chinese with English translations provided for international accessibility.

Data Processing Pipeline. Figure 3 illustrates our comprehensive data processing pipeline. Raw multimodal streams were first synchronized and segmented into 5-minute clips. We generated transcripts using Whisper v3 with speaker diarization, followed by human verification. For visual-audio captions, human annotators provided dense descriptions of each 5-minute segment played at 0.8× speed, which were then merged and refined by GPT-4o. This process yielded over 400K high-quality visual-audio captions enabling rich semantic understanding of the recorded activities.

3.2. EgoLifeQA Benchmark

Question Generation and Curation. To create realistic life assistant scenarios, we designed five question categories

Figure 1. **The Overview of EgoLife Project.** Our project presents the first week-long egocentric dataset and benchmark for life assistant research, featuring multimodal recordings from 6 participants over 7 days in a shared living environment.

Table 1. **Related Work for EgoLife Dataset — Overview of Egocentric Datasets.** For Modality, **V** denotes video, **G** denotes gaze, **I** denotes IMU, **3D** denotes 3D scans. The EgoLife dataset stands out for its ultra-long egocentric footage and rich interpersonal interactions.

Benchmark	Domain	Modality	#Captions	Size (hrs)	#Clips	Dur./Clip	Multiview	Interpersonal Dynamics
EPIC-KITCHENS [4]	Kitchen	**V**	20K+	100	700	8.5 min	✗	✗
Ego4D [1]	Daily Activities	**V G I 3D**	3.85M	3,670	9,645	22.8 min	✗	✗
EgoExo4D [2]	Skilled Activities	**V G I 3D**	500K+	1,286	5,035	1 to 42 min	✓	✗
EgoExoLearn [5]	Task Execution	**V G I**	—	120	432	13.4 min	✓	✗
EgoPet [6]	Animal Actions	**V G I**	—	84	6,646	45.5 sec	✗	✗
EgoLife	Daily Life	**V G I 3D**	400K+	266	6	44.3 h	✓	✓

reflecting core assistance needs: EntityLog tracks objects and their locations; EventRecall tests memory of specific events; HabitInsight evaluates understanding of behavioral patterns; RelationMap assesses knowledge of social dynamics; TaskMaster focuses on activity planning and management.

We initially generated approximately 100K candidate questions per participant using GPT-4o, then applied rigorous human filtering to select the most challenging and realistic 500 questions per participant (3,000 total). Each question includes multiple-choice options with carefully designed distractors, audio necessity flags, and crucially, a "certificate length" indicating the minimum temporal context needed for accurate answering.

Benchmark Statistics.

Figure 4 presents comprehensive statistics of EgoLifeQA. The benchmark is balanced across categories, with EventRecall (25%) and RelationMap (25%) being the largest categories. Critically, our analysis reveals that 2,003 out of 3,000 questions (67%) require more than 2 hours of context, with many requiring access to information spanning multiple days. This validates our hypothesis that practical life assistance demands ultra-long-context understanding far beyond current benchmarks.

Question Examples.

Figure 5 shows representative examples from each category. Questions range from simple factual recalls such as "What did I eat for breakfast yesterday?" to complex behavioral analysis like "How has my sleep pattern changed over the week?" and social understanding such as "Who have I spent the most time with in common areas?". The diversity and complexity of questions reflect real-world scenarios where life assistants must integrate information across extended time periods.

Evaluation Protocol. We evaluate models using accuracy across categories and certificate length ranges. The

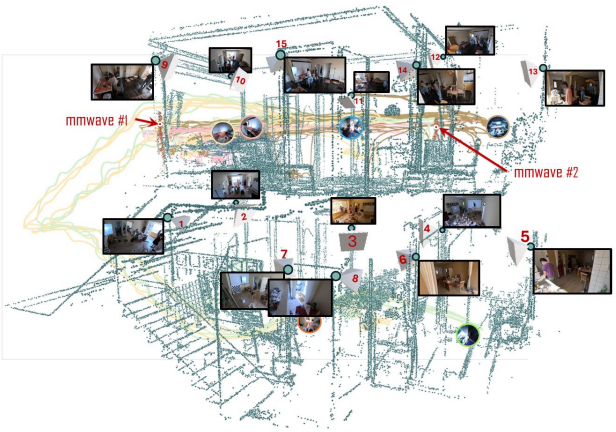


Figure 2. 3D reconstruction of the shared house using Aria Multi-MPS, showcasing the locations of 15 Exo cameras in the common area and 2 mmWave devices (highlighted in red) on the second floor.

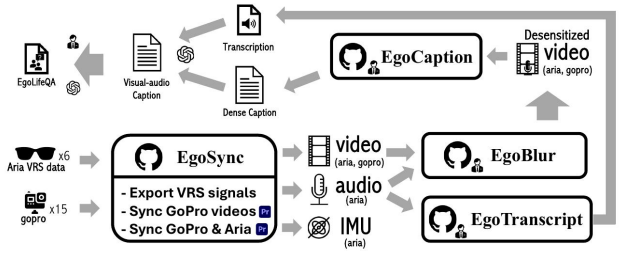


Figure 3. **The Overview of Data Process Pipeline.** Our comprehensive pipeline processes raw multimodal streams through synchronization, segmentation, transcription, and dense caption generation.

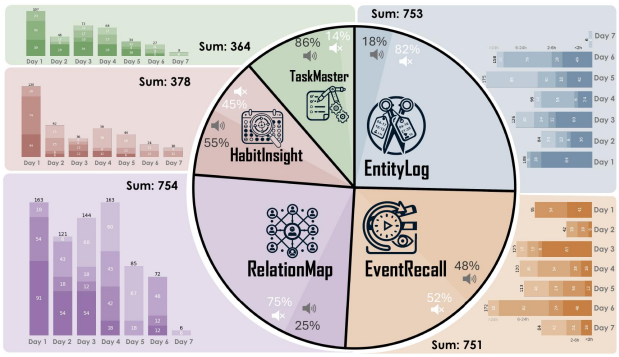


Figure 4. **EgoLifeQA Benchmark Statistics.** Comprehensive statistics showing the distribution of questions across categories and certificate length ranges.

certificate length analysis is particularly important, as it reveals how model performance degrades as longer context is required. This metric directly measures a model's capability

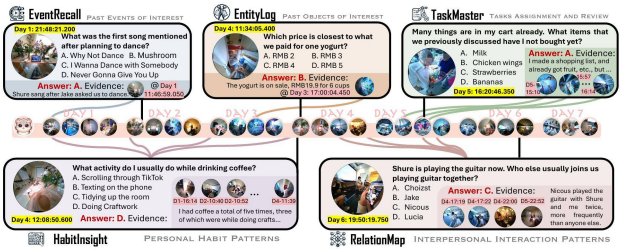


Figure 5. **EgoLifeQA Question Examples.** Representative examples from each category showing the diversity and complexity of questions in our benchmark.

for ultra-long-context reasoning essential for life assistance applications.

4. EgoButler: Agentic Egocentric Life Assistant

EgoButler addresses the dual challenges of omni-modal clip understanding and ultra-long-context reasoning through a two-stage architecture shown in Figure 6. The first stage, EgoGPT, provides dense captioning and clip-level understanding, while the second stage, EgoRAG, enables efficient retrieval and reasoning over week-long memory banks.

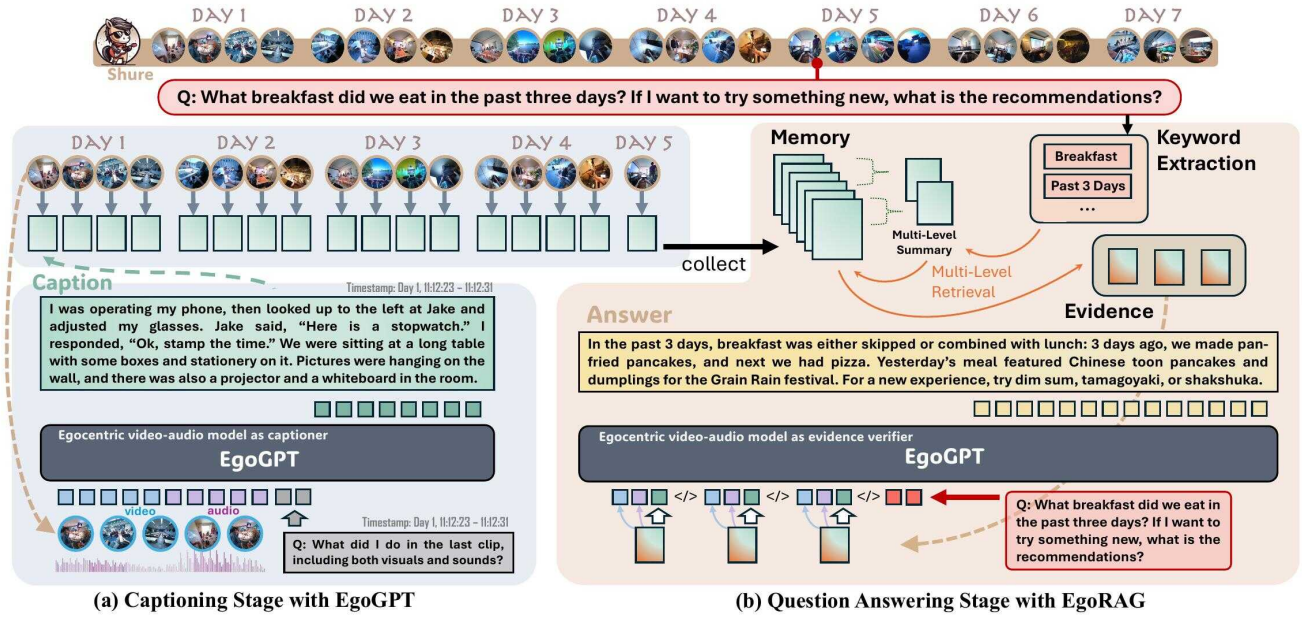
4.1. Stage 1: EgoGPT for Omni-Modal Understanding

Architecture. EgoGPT extends LLaVA-OneVision [3] with audio processing capabilities to handle the multimodal nature of egocentric data. Our model integrates three key components: (1) A visual encoder based on SigLIP for processing egocentric video frames; (2) A Whisper v3-based audio encoder for speech and environmental sound understanding; and (3) A unified language model that fuses visual and audio representations.

The audio branch processes raw audio through Whisper v3's encoder, then projects the resulting features into the language model's embedding space via a learned speech projector. Visual features from multiple frames are processed through the vision tower and resampled before projection. Both modalities are integrated through learned tokens (IMAGE_TOKEN and SPEECH_TOKEN) that enable seamless multimodal understanding.

Instruction Tuning Dataset: EgoIT-99K. To train EgoGPT for egocentric understanding, we curated EgoIT-99K, a comprehensive instruction-tuning dataset spanning 9 classic egocentric video datasets. Table 2 summarizes the dataset composition, totaling 99.48K question-answer pairs across 43.16 hours of video from diverse egocentric scenarios.

The dataset includes multiple instruction types: Video Captioning (VC), Audio-Video Captioning (AVC), Multiple Choice Questions (MCQ), Multi-Round Questions (MRC),



(a) Captioning Stage with EgoGPT **(b) Question Answering Stage with EgoRAG**

Figure 6. **The EgoButler Architecture.** Our two-stage framework combines EgoGPT for omni-modal clip understanding with EgoRAG for hierarchical memory retrieval over week-long recordings.

Table 2. **Dataset Composition of EgoIT-99K.** We curated 9 classic egocentric video datasets and leveraged their annotations to generate captioning and QA instruction-tuning data for fine-tuning EgoGPT, building on the LLaVA-OneVision base model [3]. **#AV** means the number of videos with audio used for training. QAs include multiple types — **VC**: Video Captioning, **AVC**: Audio-Video Captioning, **MCQ**: Multiple Choice Questions, **MRC**: Multi-Round Questions, **IQA**: Image Question-Answering.

Dataset	Duration	#Videos (#AV)	#QA	QA Type
Ego4D [1]	3.34h	523 (458)	1.41K	VC, AVC, MCQ, MRC
Charades-Ego [12]	5.04h	591 (228)	18.46K	VC, AVC, MRC
HoloAssist [13]	9.17h	121	33.96K	VC, MCQ, MRC, IQA
EGTEA Gaze+ [14]	3.01h	16	11.20K	VC, MCQ, MRC, IQA
IndustReal [15]	2.96h	44	11.58K	VC, MCQ, MRC, IQA
EgoTaskQA [16]	8.72h	172	3.59K	VC, MCQ, MRC
EgoProceL [17]	3.11h	18	5.90K	VC, MCQ, MRC, IQA
Epic-Kitchens [4]	4.15h	36	10.15K	VC, MCQ, MRC, IQA
ADL [18]	3.66h	8	3.23K	VC, MCQ, MRC, IQA
Total	**43.16h**	**1529 (686)**	**99.48K**	

and Image Question-Answering (IQA). This diversity ensures robust understanding across different egocentric tasks and modalities. Notably, 686 out of 1,529 videos include audio annotations, enabling effective audio-visual learning.

Personalization via Day-1 Learning. A key insight from our experiments is that personalization significantly improves life assistant performance. We implement identity-aware learning by fine-tuning EgoGPT on Day-1 data from each participant, allowing the model to learn individual habits, preferences, and behavioral patterns. This personalization proves particularly effective for RelationMap and HabitInsight tasks requiring understanding of individual social dynamics and routines.

4.2. Stage 2: EgoRAG for Long-Context Reasoning

Hierarchical Memory Architecture. EgoRAG addresses the fundamental challenge of reasoning over week-long recordings through hierarchical memory organization. We construct three levels of memory banks: (1) Clip-level captions (5-minute segments); (2) Hour-level summaries aggregating clip information; and (3) Day-level summaries capturing daily patterns and significant events.

This hierarchical structure enables efficient retrieval at different temporal granularities. For questions requiring recent context, clip-level retrieval suffices. For longer-term queries, hour and day-level summaries provide compressed but relevant information while maintaining computational efficiency.

Retrieval and Answer Generation. Given a question, EgoRAG first determines the appropriate temporal scope based on question analysis. It then retrieves the top-k most relevant memory segments using semantic similarity search implemented via ChromaDB embeddings. The retrieved segments serve as context for EgoGPT to generate the final answer, effectively combining precise clip-level understanding with long-term memory access.

Our retrieval strategy adaptively selects memory levels based on the question's certificate length and temporal scope. This approach ensures that models have access to the most relevant information while avoiding the computational overhead of processing entire week-long recordings for every query.

Table 3. **Performance of EgoGPT.** The table compares EgoGPT with state-of-the-art commercial and open-source models on existing egocentric benchmarks.

Model	#Param	#Frames	EgoSchema	EgoPlan	EgoThink
GPT-4v [7]	—	32	56.6	38.0	65.5
Gemini-1.5-Pro [8]	—	32	72.2	31.3	62.4
GPT-4o [19]	—	32	72.2	32.8	65.5
LLaVA-Next-Video [10]	7B	32	49.7	29.0	40.6
LongVA [11]	7B	32	44.1	29.9	48.3
IXC-2.5 [20]	7B	32	54.6	29.4	56.0
InternVideo2 [21]	8B	32	55.2	27.5	43.9
Qwen2-VL [9]	7B	32	66.7	34.3	59.3
Oryx [22]	7B	32	56.0	33.2	53.1
LLaVA-OV [3]	7B	32	60.1	30.7	54.2
LLaVA-Videos [23]	7B	32	57.3	33.6	56.4
EgoGPT (EgoIT)	7B	32	73.2	32.4	61.7
EgoGPT (EgoIT+EgoLifeD1)	7B	32	75.4	33.4	61.4

5. Experiments

5.1. Experimental Setup

Baselines and Evaluation. We evaluate EgoButler against state-of-the-art commercial and open-source multimodal models. Commercial baselines include GPT-4V [7], Gemini-1.5-Pro [8], and GPT-4o [19]. Open-source baselines include LLaVA-OneVision [3], LLaVA-Next-Video [10], Qwen2-VL [9], and other recent video-language models. For fair comparison on EgoLifeQA, we integrate each baseline model into our EgoRAG framework, replacing only the captioning component.

Implementation Details. EgoGPT is implemented as a 7B parameter model built on LLaVA-OneVision with additional audio processing capabilities. We use 32 frames for video processing and 1 FPS sampling for caption generation. The model is trained on EgoIT-99K for 3 epochs with learning rate 2e-5. Day-1 personalization uses an additional 1 epoch of training on participant-specific data.

5.2. Results on Existing Egocentric Benchmarks

Table 3 shows EgoGPT's performance on three established egocentric benchmarks: EgoSchema, EgoPlan, and EgoThink. Our model achieves state-of-the-art results across all benchmarks, demonstrating the effectiveness of our omni-modal architecture and instruction tuning strategy.

EgoGPT (EgoIT-99K) achieves 73.2 on EgoSchema, 32.4 on EgoPlan, and 61.7 on EgoThink, outperforming open-source models and reaching competitive performance with commercial models. The addition of Day-1 personalization (EgoGPT EgoIT+EgoLifeD1) further improves performance to 75.4, 33.4, and 61.4 respectively, setting new state-of-the-art results for open-source models.

These results validate our approach of combining comprehensive instruction tuning with personalization, showing that models can effectively transfer knowledge from diverse egocentric scenarios to new benchmark tasks.

5.3. Results on EgoLifeQA Benchmark

Table 4 presents comprehensive results on our EgoLifeQA benchmark. Commercial models achieve competitive performance, with Gemini-1.5-Pro reaching 36.9 average accuracy and GPT-4o achieving 36.2. However, these models struggle with certain categories, particularly RelationMap tasks requiring understanding of social dynamics.

EgoGPT without personalization achieves 33.1 average accuracy, demonstrating reasonable performance but trailing commercial baselines. The addition of Day-1 personalization significantly improves performance to 36.0, matching commercial model performance and achieving notable gains in RelationMap (29.6 $\rightarrow$ 33.6) and EntityLog (35.2 $\rightarrow$ 39.2) categories.

Impact of Personalization. The consistent improvement from personalization across all categories highlights its importance for life assistant applications. Personalization helps models understand individual behavioral patterns, social relationships, and environmental contexts that are crucial for accurate assistance.

5.4. Effectiveness of EgoRAG

Table 5 demonstrates the critical importance of our EgoRAG component for long-context reasoning. We evaluate models across different certificate length ranges to assess their capability for ultra-long-context understanding.

For questions requiring less than 2 hours of context, all models achieve similar performance (27–28 accuracy). However, as certificate length increases, the gap widens dramatically. Without retrieval, both Gemini-1.5-Pro and EgoGPT show degraded performance on longer certificate lengths, dropping to 18.4 and 25.0 respectively for $>$24h questions.

EgoRAG provides substantial improvements, particularly for long-context questions: 6–24h certificate length improves from 26.8 to 38.9, and $>$24h improves from 25.0 to 35.4. These results validate our hypothesis that hierarchical retrieval is essential for practical life assistant applications requiring ultra-long-context reasoning.

5.5. Ablation Studies

Caption Quality Analysis. Table 6 reveals the critical importance of high-quality multimodal captions for EgoButler performance. Human visual-audio captions achieve the highest performance (45.5), significantly outperforming both human narrations (31.5) and transcripts (29.6). This finding emphasizes the value of our dense visual-audio captioning process.

Among automated captioning methods, EgoGPT models with both visual and audio modalities perform best. Audio-only models achieve 27.2–28.1 accuracy, while visual-only models reach 31.2–33.6. The combination of visual and audio modalities yields 33.1–36.0 accuracy, with personal-

Table 4. **Performance comparison of EgoGPT with state-of-the-art models on EgoLifeQA benchmarks.** For a fair comparison on EgoLifeQA, EgoGPT was replaced with the corresponding models in the EgoButler pipeline to evaluate their performance under the same conditions. Models that provide captions for EgoLifeQA use 1 FPS for video sampling.

Model	#Frames	Audio	Identity	EgoLifeQA					
				EntityLog	EventRecall	HabitInsight	RelationMap	TaskMaster	Average
Gemini-1.5-Pro [8]	—	✓	✗	36.0	37.3	45.9	30.4	34.9	36.9
GPT-4o [19]	1 FPS	✗	✗	34.4	42.1	29.5	30.4	44.4	36.2
LLaVA-OV [3]	1 FPS	✗	✗	36.8	34.9	31.1	22.4	28.6	30.8
EgoGPT (EgoIT-99K)	1 FPS	✓	✗	35.2	36.5	27.9	29.6	36.5	33.1
EgoGPT (EgoIT-99K+D1)	1 FPS	✓	✓	39.2	36.5	31.1	33.6	39.7	36.0

Table 5. **Effectiveness of EgoRAG.** Integrating EgoRAG significantly enhances video-language models' performance in long-context question answering, especially for questions requiring longer certification lengths. For comparison, we evaluate Gemini-1.5-Pro and EgoGPT on a half-hour video segment, limiting their answers to this timeframe.

Model	Certificate Length			
	$< 2h$	$2h - 6h$	$6h - 24h$	$> 24h$
Gemini-1.5-Pro	27.9	14.8	25.0	18.4
EgoGPT	28.2	29.1	26.8	25.0
EgoGPT+EgoRAG	27.2	35.7	38.9	35.4

Table 6. **Ablation Study on EgoGPT.** We construct different EgoRAG memory banks using generated captions from EgoGPT variants. The first three rows use captions from human annotations as a reference. All response generation models utilize EgoGPT (EgoIT-99K+D1) to ensure fair comparison. The result indicates how caption quality affects of EgoBulter performance.

Caption Source	Visual	Audio	Dataset	Avg.
Narration	✓	✓	—	31.5
Transcript	✗	✓	-	29.6
Visual-Audio Caption	✓	✓	—	45.5
EgoGPT (Audio Only)	✗	✓	EgoIT-99K	27.2
EgoGPT (Audio Only)	✗	✓	EgoIT-99K+D1	28.1
EgoGPT (Visual Only)	✓	✗	EgoIT-99K	31.2
EgoGPT (Visual Only)	✓	✗	EgoIT-99K+D1	33.6
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K	33.1
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K+D1	36.0

ization providing consistent improvements across all configurations.

Modality Importance. Our experiments demonstrate that both visual and audio modalities are essential for egocentric understanding. Audio provides crucial context for conversations, environmental sounds, and activities not visible in video, while visual information captures spatial relationships, object interactions, and non-verbal behaviors. The synergy between modalities proves particularly important for RelationMap and HabitInsight categories.

5.6. Qualitative Analysis

Figure 7 shows qualitative comparisons between EgoGPT and Gemini-1.5-Pro under the EgoButler framework. Our model demonstrates superior understanding of temporal relationships and interpersonal dynamics, correctly identifying complex social patterns that commercial models miss.

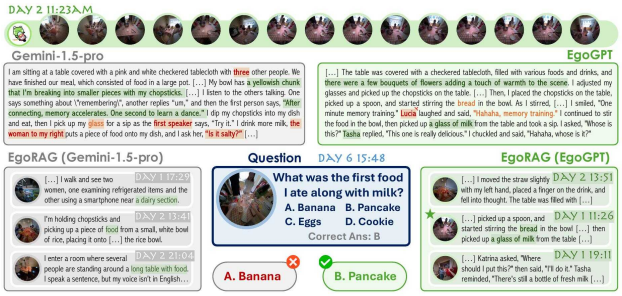


Figure 7. **Qualitative Analysis.** Comparisons between EgoGPT and Gemini-1.5-Pro showing superior understanding of temporal relationships and interpersonal dynamics.

The examples highlight EgoButler's ability to integrate information across multiple days, track object locations over time, and understand behavioral patterns that emerge from long-term observation. These capabilities are essential for practical life assistant applications requiring deep understanding of personal contexts and routines.

6. Conclusion and Outlook

We have presented EgoLife, the first week-long multimodal egocentric dataset specifically designed for life assistant research, along with EgoLifeQA, a comprehensive benchmark highlighting the importance of ultra-long-context reasoning. Our EgoButler framework demonstrates that combining omni-modal understanding through EgoGPT with hierarchical memory retrieval through EgoRAG enables significant advances in egocentric life assistant capabilities.

Key findings from our work include: (1) Ultra-long context is essential for practical life assistance, with 67% of realistic questions requiring more than 2 hours of temporal context; (2) Personalization provides substantial improvements across all assistance categories, particularly for social understanding and habit analysis; (3) High-quality multimodal captions are crucial, with human visual-audio descriptions outperforming automated alternatives; (4) Hierarchical retrieval architecture enables efficient access to week-long memories while maintaining computational tractability.

EgoGPT achieves state-of-the-art performance on existing egocentric benchmarks while EgoRAG provides dramatic improvements for long-context reasoning tasks. These advances represent a significant step toward practical egocentric life assistants capable of understanding and supporting human activities over extended time periods.

Limitations and Future Work. While EgoLife provides valuable insights into long-term egocentric understanding, our dataset is limited to 6 participants and 7 days in a single environment. Future work should explore larger-scale collections across diverse populations and environments. Additionally, our current approach focuses on reactive assistance based on historical data; incorporating predictive capabilities for proactive assistance represents an important research direction.

Privacy and ethical considerations remain paramount as egocentric life assistants become more capable. Future research must address data minimization, on-device processing, and user control over personal information while maintaining assistance quality.

Broader Impact. Egocentric life assistants have the potential to significantly improve quality of life for individuals with memory challenges, busy professionals, and elderly populations. However, the intimate nature of egocentric data raises important privacy concerns that must be carefully addressed through technical and policy measures. Our work provides a foundation for responsible development of such systems while highlighting key research challenges that remain to be solved.

References

[1] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, et al. Ego4d: Around the world in 3,000 hours of egocentric video. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 18995–19012, June 2022. 1, 2, 3, 5

[2] Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, et al. Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 19383–19400, June 2024. 1, 2, 3

[3] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task transfer. arXiv preprint arXiv:2408.03326, 2024. 1, 2, 4, 5, 6, 7

[4] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, et al. The epic-kitchens dataset: Collection, challenges and baselines. IEEE Transactions on Pattern Analysis and Machine Intelligence, 43(11):4125–4141, 2021. 2, 3, 5

[5] Yifei Huang, Guo Chen, Jilan Xu, Mingfang Zhang, Lijin Yang, Baoqi Pei, Hongjie Zhang, Dong Lu, Yali Wang, et al. Egoexolearn: A dataset for bridging asynchronous ego- and exo-centric view of procedural activities in real world. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024. 3

[6] Amir Bar, Arya Bakhtiar, Danny Tran, Antonio Loquercio, Jathushan Rajasegaran, Yann LeCun, Amir Globerson, and Trevor Darrell. Egopet: Egomotion and interaction data from an animal's perspective. arXiv preprint arXiv:2404.09991, 2024. 3

[7] OpenAI. Gpt-4v(ision) system card, 2023. 2, 6

[8] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. arXiv preprint arXiv:2312.11805, 2023. 2, 6, 7

[9] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, et al. Qwen2-audio technical report. arXiv preprint arXiv:2407.10759, 2024. 2, 6

[10] Yuanhan Zhang, Bo Li, haotian Liu, Yong jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model, April 2024. 2, 6

[11] Peiyuan Zhang, Kaichen Zhang, Bo Li, Guangtao Zeng, Jingkang Yang, Yuanhan Zhang, Ziyue Wang, Haoran Tan, Chunyuan Li, and Ziwei Liu. Long context transfer from language to vision. arXiv preprint arXiv:2406.16852, 2024. 2, 6

[12] Gunnar A. Sigurdsson, Abhinav Gupta, Cordelia Schmid, Ali Farhadi, and Karteek Alahari. Charades-ego: A large-scale dataset of paired third and first person videos, 2018. 5

[13] Xin Wang, Taein Kwon, Mahdi Rad, Bowen Pan, Ishani Chakraborty, Sean Andrist, Dan Bohus, Ashley Feniello, Bugra Tekin, Felipe Vieira Frujeri, et al. Holoassist: an egocentric human interaction dataset for interactive ai assistants in the real world. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 20270–20281, 2023. 5

[14] Yin Li, Miao Liu, and James M. Rehg. In the eye of beholder: Joint learning of gaze and actions in first person video. In European Conference on Computer Vision (ECCV), 2018. 5

[15] Tim J Schoonbeek, Tim Houben, Hans Onvlee, Fons van der Sommen, et al. Industreal: A dataset for procedure step recognition handling execution errors in egocentric videos in an industrial-like setting. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pages 4365–4374, 2024. 5

[16] Baoxiong Jia, Ting Lei, Song-Chun Zhu, and Siyuan Huang. Egotaskqa: Understanding human tasks in egocentric videos. Advances in Neural Information Processing Systems, 35:3343–3360, 2022. 5

[17] Siddhant Bansal, Chetan Arora, and C.V. Jawahar. My view is the best view: Procedure learning from egocentric videos. In European Conference on Computer Vision (ECCV), 2022. 5

[18] Junhao Pan, Zehua Yuan, Xiaofan Zhang, and Deming Chen. Youhome system and dataset: Making your home know you better. IEEE International Symposium on Smart Electronic Systems (IEEE - iSES), 2022. 5

[19] OpenAI. Gpt-4o system card, 2024. 6, 7

[20] Pan Zhang, Xiaoyi Dong, Bin Wang, Yuhang Cao, Chao Xu, Linke Ouyang, Zhiyuan Zhao, Haodong Duan, Songyang Zhang, Shuangrui Ding, et al. Internlm-xcomposer: A vision-language large model for advanced text-image comprehension and composition. arXiv preprint arXiv:2309.15112, 2023. 6

[21] Yi Wang, Kunchang Li, Xinhao Li, Jiashuo Yu, Yinan He, Guo Chen, Baoqi Pei, Rongkun Zheng, Jilan Xu, Zun Wang, et al. Internvideo2: Scaling video foundation models for multimodal video understanding. arXiv preprint arXiv:2403.15377, 2024. 6

[22] Zuyan Liu, Yuhao Dong, Ziwei Liu, Winston Hu, Jiwen Lu, and Yongming Rao. Oryx mllm: On-demand spatial-temporal understanding at arbitrary resolution. arXiv preprint arXiv:2409.12961, 2024. 6

[23] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video instruction tuning with synthetic data. arXiv preprint arXiv:2410.02713, 2024. 6